R.D. DUTTA & CO. PVT. LTD.

302, Rattan Jyoti Building 18,
Rajendra Place, New Delhi - 110008
Phone : 25710739, 25739805
Email : info@rddutta.com
Web : www.rddutta.com
CIN No. : U74899DL1982PTC014209

Quotation No : Q24000516

Date : 08-Oct-2024

KEI Industries Ltd.
A-280-284, RIICO Industrial Area, Chopanki
Bhiwadi
RAJASTHAN

Kind Attention : Mr./Ms. D.N.Sharma

Dear Sir,

It's been more than 30 years since R D Dutta & Company came into existence in the field of Compressed air systems and accessories.

We have an expertise in Providing best of sales and after sales support and a wide range of products to meet our customer needs.

To us being leader not only means selling the best quality products but, it also means working hard to understand your industry, your needs and the demands placed on your productivity in order to provide you with the most beneficial solutions for your work.
We trust you Find our offer in line with your requirement.

Should you require any further clarification/information, please contact the undersigned.
Thanking you of out best attention at all times.

Very truly yours,

For R. D. Dutta & Co. PVT. LTD.

Rakesh Dutta

Director

9311955930

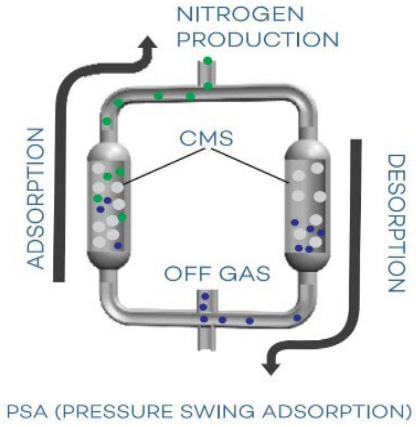
DT1005

nGEN POWERED BY ABSSTEM INDUSTRIAL

Industrial PSA Nitrogen Generator



PSA Technology



PSA (PRESSURE SWING ADSORPTION)

Industrial Nitrogen Generator mainly consists of two adsorbers filled with carbon molecular sieves. Pressurized air is passed through one of these adsorbers, carbon having the property to attract oxygen, adsorbs oxygen from air and Nitrogen is allowed to pass through and move into the storage tank as the final product. As the adsorber approaches saturation, the air is allowed to pass through the other adsorber while this adsorber is being regenerated to ensure a continuous supply of Nitrogen. This changeover takes place through a specially designed microcontroller to ensure a hassle-free operation. The quality of Nitrogen produced is monitored by a Nitrogen analyser.

Key Features

- Shipped ready for installation
- 5 minutes to run and never have to stop it!
- Easy installation
- Low operating expenses compared to gas cylinders/liquid Nitrogen
- Maintenance costs are very less
- Embedded Safety Interlocks
- Space-saving design
- Eliminates all hazards which are there with gas cylinder or liquid Nitrogen
- Longevity: at least 10 years

Absstem Industrial Solutions Pvt. Ltd.
Plot No. 591, Sector 8, Phase IV
IMT Manesar, Gurugram, Haryana - 122051
Contact - 1800 3010 3394 Email - indsales@absstem.com
CIN: U29256HR2022PTC103387

Description

a.	Product Name	Nitrogen Gas Generator
b.	Model Range	**NGEN**
	Technology	PSA (Pressure Swing Adsorption)

Specifications

a.	Nitrogen Flow Rate	As per Performance Sheet
b.	Nitrogen Purity	As per Performance Sheet
c.	Nitrogen Delivery Pressure	5.5 Barg (Higher Pressure Options Available)
d.	Ambient conditions	
	Temperature	0 to 50 Deg. Celsius
	Relative Humidity	up to 80 % (Max.)
	Altitude	up to 1000 m above Sea level

Scheme Offered – (PSA Only)

This scheme will comprise of

- Air package with inlet air filtration system
  (Air compressor, Air dryer, Water Separator, Oil filters)
- Industrial Nitrogen generator
- Storage Tank
- Nitrogen Analyzer
- Alarm panel

Unique Features of Absstem

a. All the fabricated items are cleaned and degreased following our strict quality policy.
b. All the Instruments are specially manufactured to make them suitable for Nitrogen gas.
c. Guaranteed 10+ years of healthy life.
d. We provide the purest **Nitrogen** in Industry.
e. Data logging and Graphical Reports for all parameters available.
f. Test certificates of all instruments are available on demand.

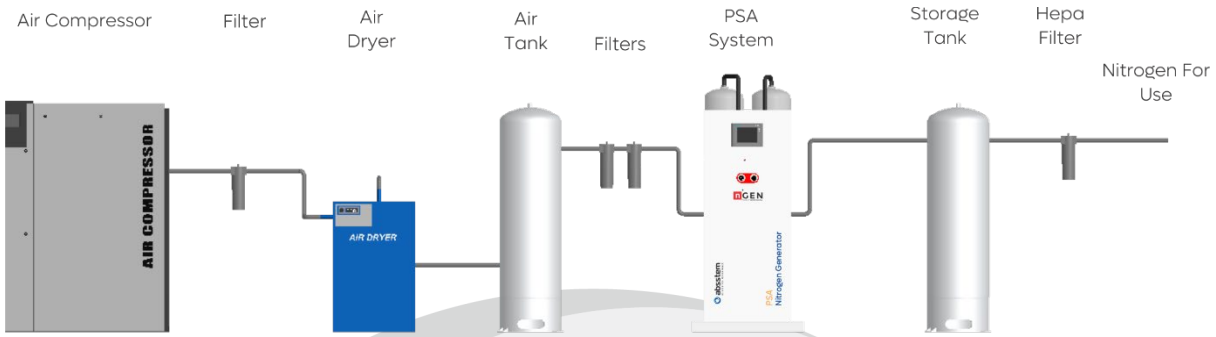
Onsite Gas Generation vs Liquid Tank/Cylinder Alternative

Non-stop availability: 24 hours a day, 7 days a week
Easy integration within existing installations
Your own independent supply of industrial gas
Significant economies for scale and lower operational costs: no rental charges, logistical expenses and bulk user evaporation losses
No safety hazards when handling high-pressure cylinders

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Equipment Description

The Industrial Nitrogen Generator consists of



Air Compressor	An Air compressor is used to compress the atmospheric air to a pressure required for the separation of **Nitrogen** from the air.
Refrigeration dryer	Compressed air then passes through a refrigerated dryer to remove moisture (which is harmful to sieves) from the air.
Air Tank	Dry air is collected in an air tank to remove the pulsations which serve the dynamic (changing) requirement of air by the PSA system.
Multistage Filtration system	Before air enters the PSA system, it is passed through multistage filtration removing moisture, particulates and oil content to the specified levels.
PSA system	Pressurized air is passed through PSA Towers filled with Molecular sieves to separate **Nitrogen** from Air.
Storage Tank	Industrial Grade **Nitrogen** produced from NGEN is stored in this tank to supply a continuous flow of **Nitrogen**.
Nitrogen Analyzer	Nitrogen produced is monitored by a **Nitrogen** analyser. In case, O2 purity goes below the specified level, it gives an alarm and switches it to vent mode.

HD Touch Screen Display

(User-friendly touchscreen interface for easy operation and maintenance)

Absstem industrial Nitrogen generator complies with all industry standards and regulation

- Alarm management system
- Plug and play
- Fully automatic
- Process data display
- Consumables running hours display
- Data logging via USB drive
- Graphical representation



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Additional upgrades

- Digital flow meter with totaliser
- High-pressure Nitrogen tank
- Cylinder manifold and alarm management system
- Remote monitoring

Plant Operation

- Unit operates 24-hrs a day, 7-days a week reliably, automatically & safely

Automation

- The Plant is fully automatic and needs only electrical power for running the Air compressor
- The machine automatically enters into standby mode when the level in the Storage Tank reaches its set limit. This helps to increase the life of the Air compressor.

International Quality standards

A strict quality procedure, following the leading international pharmacopoeia and Industrial directives, is followed in the selection of all mechanical and electrical instruments, vessel fabrication and designing of Industrial Nitrogen generators.

Itemised Supply Description

No.	Item description	Make	Qty
a	Air Compressor with Filters	ABSSTEM APPROVED	Quoted Separately
b	Refrigeration Dryer	ABSSTEM APPROVED	Quoted Separately
c	Air Tank	ABSSTEM	Quoted Separately
d	Set of Filters	ABSSTEM	1 no.
e	PSA unit (NGEN)	ABSSTEM	1 no.
f	Electrical Power Panel with PLC and all cabling	ABSSTEM & MITSUBISHI	1 no.
f	Digital Nitrogen Analyzer	ABSSTEM	1 no.
g	Storage Tank	ABSSTEM	Quoted Separately
h	Outlet product gas Filter	ABSSTEM	1 no.

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Electrical Protection

The client should make proper arrangements for earthing/grounding systems for the Industrial **Nitrogen** Generator at their facility.

Enclosure Panel

- Ergonomically designed metal enclosure with powder-coated finish for longer life.
- High-strength IP-rated panel fittings from Germany for a robust enclosure.
- Fan filters from **Rittal** are IP-55 rated making the electrical enclosure's internal environment dust-free.
- Easy connectivity to any place of installation with req. port connections.
- High-strength base frame designed for higher load and easy transportation.
- Vessels are designed per ASME section VIII DIV I. Vessels are epoxy paint coated which is considered good for longevity and corrosion resistance.
- All electrical items are CE certified with operating voltage at 24 VDC

Annual Maintenance Contract (AMC)

Our in-house support Absstem shield is 24*7 open and we assure a timely service maintenance schedule. Per the schedule mentioned in the operational manual, we suggest the operator change the filters by following a simple protocol.

It is very easy to change or replace as our commissioning engineer plus our commissioning engineer and tech support office will always guide you about the activity.

Best ROI Guaranteed

ROI within 6 to 18 months when compared to conventional methods

Generator performance depends on the component selection. Our processes make this job easier as all equipment is selected based on more than 3 decades of research. NGEN is a fully automatic (operator-free operation) that ensures peace of mind.

We deliver what we commit!!

Thank You

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Product Description

Product	Description	Make	Qty
nGen Air Package	nGEN 20 CP / nGenx 10 CP / nGen 13 CP / nGen 15 CP Nitrogen		2

Caption	Specification
Air Compressor KW	15 KW
Air Compressor Make	Atlas Copco/ Ingersoll Rand/ Gardner Denver/ As per Demand
Air Dryer	Included
Dry Air Storage	Included
Interconnecting Piping	Included
Pre filter	Included
Dry Air Generated	101 CFM

Product	Description	Make	Qty
nGen Nitrogen Generator	nGEN 15 Nitrogen		2

Caption	Specification
Nitrogen	Gas Generated
Gas Pressure	upto 6.5 Barg(Higher options Available)
Flow @ 95 %	66 Nm3/hr
Flow @ 97 %	60 Nm3/hr
Flow @ 98 %	54 Nm3/hr
Flow @ 99 %	50 Nm3/hr
Flow @ 99.5 %	44 Nm3/hr
Flow @ 99.9 %	40 Nm3/hr
Flow @ 99.95 %	33 Nm3/hr
Compressed Air Requirement	89 CFM
Module	PSA Plant
Air Compressor And Drier	In your Scope / Quoted Separately

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Product	Description	Make	Qty
nGen Gas Storage	nGEN 15 / 20 ST Nitrogen		2

Caption	Specification
Water Volume	3000 Liters
Gas Stored	Nitrogen
Gas Stored Volume	19.5 M3
Working Pressure	6.5 Barg
Hydrotest Pressure	9 Barg

Product	Description	Make	Qty
Lubricated Booster	IR-7T2X12.5 Booster		2

Caption	Specification
Gas to be compressed	Nitrogen
Suction Pressure	3 to 4 Barg
Discharge pressure	8 to 35 Barg
Motor rating	10 KW
Voltage	415 V AC
Gas Flow	25 to 37 NM3/hr
Type	Air Cooled Lubricated Reciprocating Booster
Accessories Included	Pressure Regulating System, Suction Pot, Pressure transmitter and pressure gauge at suction pot
After Cooler	Electric Motor Driven Finned type After Cooler
After Cooler Power	0.7 KW, 3 Phase
After Cooler Capacity	100 m3/hr

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Product	Description	Make	Qty
HP Storage Tank 20 Barg	6000 ltr HP Storage Tank 20 Barg High Pressure Storage Tank	ABSSTEM	2

Caption	Specification
Capacity	6000 litres
Working Pressure	20 Barg
Hydrotest Pressure	33 Barg
Shell / Dish Thickness	16 mm / 18 mm
In / Out Connection	80 NB
Design Standard	IS 2825

Commercial Offer

S No	Item Code	Item Name	Description	HSN	Qty	Unit Price	GST(%)	Amount
1	nGEN 20 CP	nGEN 20 CP / nGenx 10 CP / nGen 13 CP / nGen 15 CP	nGen Air Package	840510	2	10,02,000.00	18.00	20,04,000.00
2	nGEN 15	nGEN 15	nGen Nitrogen Generator	840510	2	12,51,600.00	18.00	25,03,200.00
3	nGEN 20 ST	nGEN 15 / 20 ST	nGen Gas Storage	840510	2	2,52,000.00	18.00	5,04,000.00
4	IR-7T2X12.5	IR-7T2X12.5	Lubricated Booster	8414	2	3,90,000.00	18.00	7,80,000.00
5	HPS1	6000 ltr HP Storage Tank 20 Barg	HP Storage Tank 20 Barg	73110090	2	8,07,840.00	18.00	16,15,680.00

Sub Total	74,06,880.00
GST(%) @18.00	13,33,238.40
Grand Total	87,40,118.40

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Terms And Conditions

1.	EXCLUSIONS & CLIENTS SCOPE	All erection / civil work, All Power Cables & Cabling out of battery limit of PSA Plant, Cable Trenches etc. Unloading / Shifting of Gas Plant at Clients site will be in Client Scope. Additional gas piping, cooling water (if required).
2.	SUPERVISION OF ERECTION & COMMISSIONING	There shall be no commissioning charges, however you shall arrange freeTo & Fro Air Ticket, lodging / boarding, and local conveyance for our Engineers during supervision of Installation of Gas Plant at site on free of cost basis. Our Engineers will be there for Supervision only. Any manpower / fitter required for installation work shall be provided by you.
3.	P&F	3% Extra.
4.	Freight	Extra at Actuals.
5.	Insurance	In your Scope.
6.	Payment Terms	50% Advance with your Prestegious Purchase Order and Balance against PI.
7.	Delivery	6-8 Weeks from the PO and Advance. Rest can be confirmed for Availability.
8.	Warranty	We hereby confirm that, our equipment shall have Suppliers warranty for a period of 12 months from the date of Commissioning or 18 months from date of Dispatch whichever is earlier.
9.	GST	As per the Commercial Table.
10.	Remarks	.

Hope you will find our offer in line with your requirement.

Should you require any further clarification/information, please feel free to contact us.

Thanking you, and assuring you of our best attention at all times.

Very Yours truly,

For R. D. Dutta & Co. PVT. LTD.

Rakesh Dutta

Director

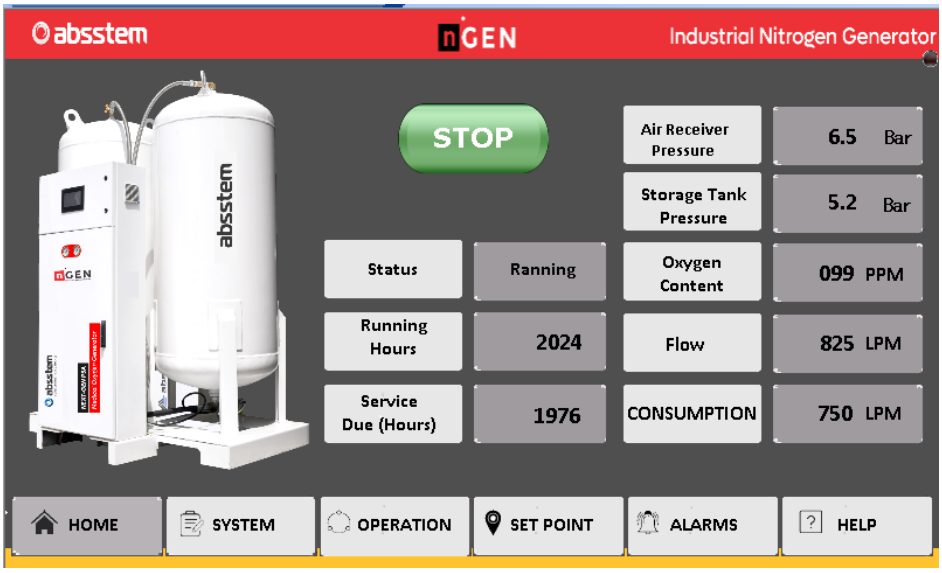
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DT1005

General Features of ‘Eye Touch Screen’

This document outlines the general features of ‘Eye touch’ screen.

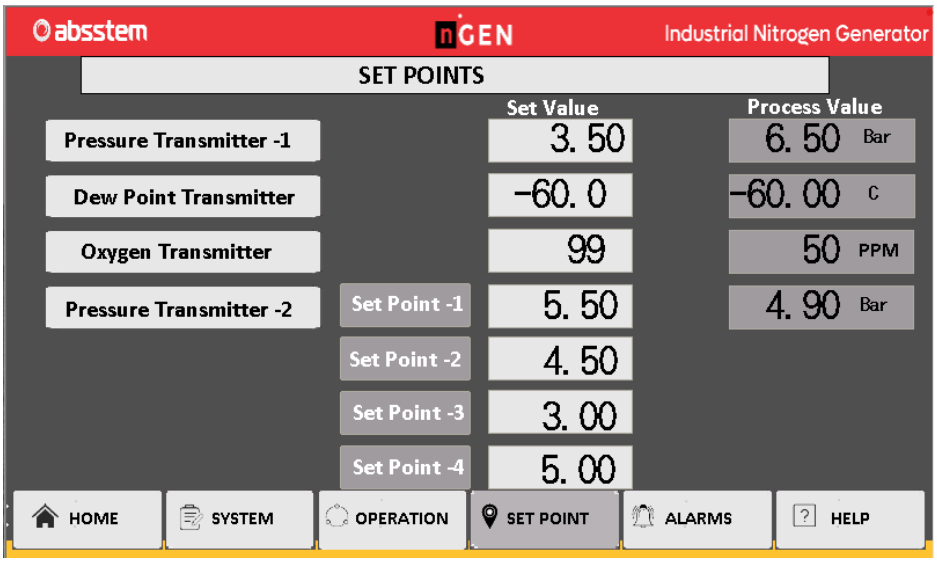
Home Page:



- **Real-time data visualization:**
  - Running/stop status of the machine
  - Live storage pressure
  - Total Running hours
  - Service due hours
  - Storage gas purity (%,PPM)
  - Flow
  - Consumption
  - Live inlet pressure

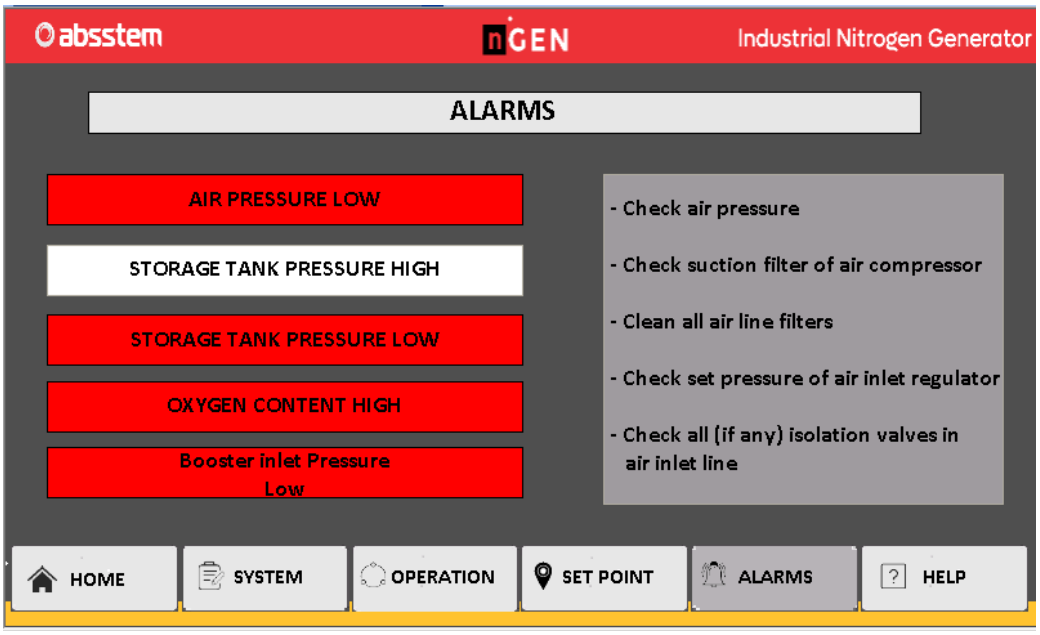
- **Intuitive interface:** Large, clearly labeled buttons and text for easy navigation and interaction.

- **Adjustable Settings:**

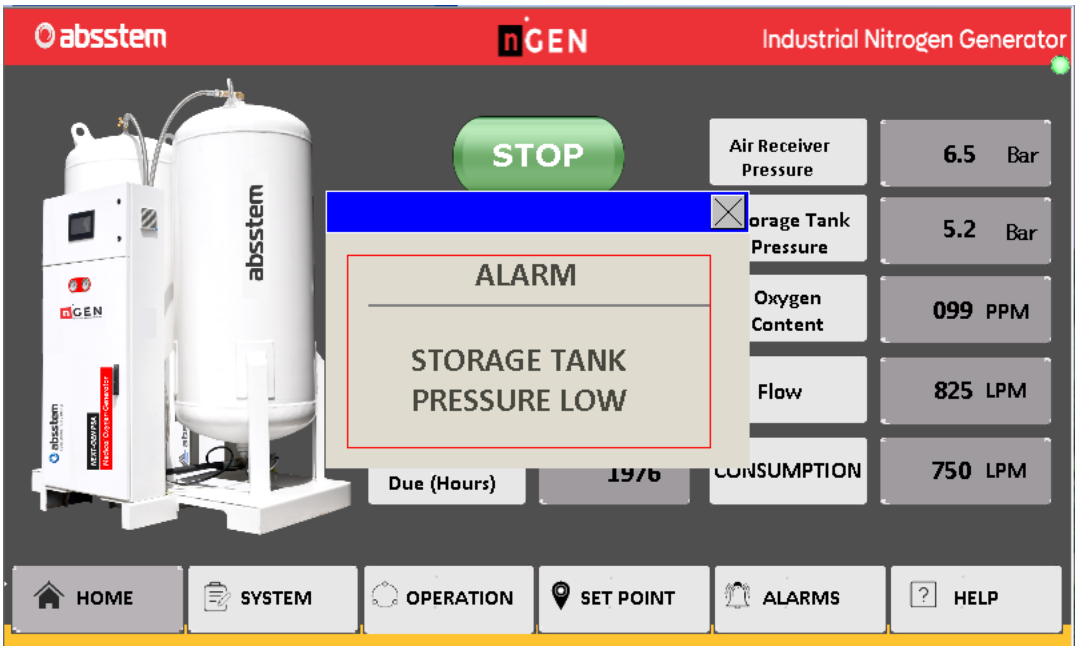


- **Inlet pressure on/off interlock:** Set point for automatic start/stop of machine based on inlet pressure.
- **Dew Point Interlock:** Set the limit of dew point as needed.
- **Storage max pressure limit:** Define the maximum safe pressure for the storage tank.
- **Storage pressure alarm control adjust point:** Set the threshold for triggering a low-pressure alarm.
- **Purity adjusts point:** Set the threshold for triggering a low-purity alarm.
- **Idle Mode:** Set the point of energy saving mode.
- **Auto start point:** Plant start automatic from energy saving mode when use reach the adjust point.

Alarm System:

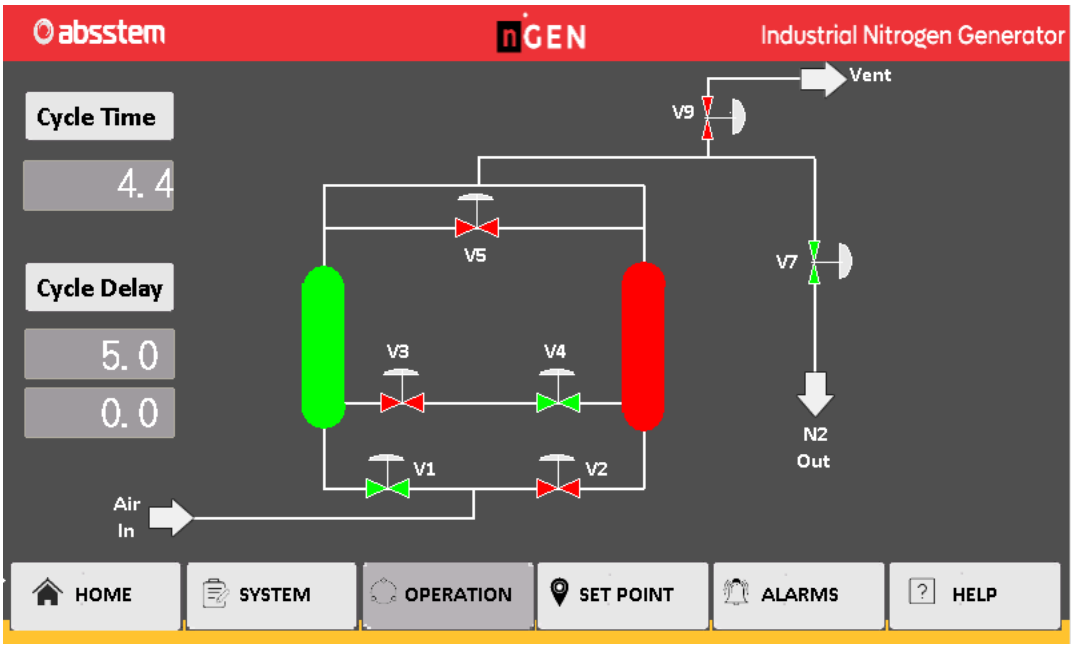


- **Popup notifications:** Alerts for:



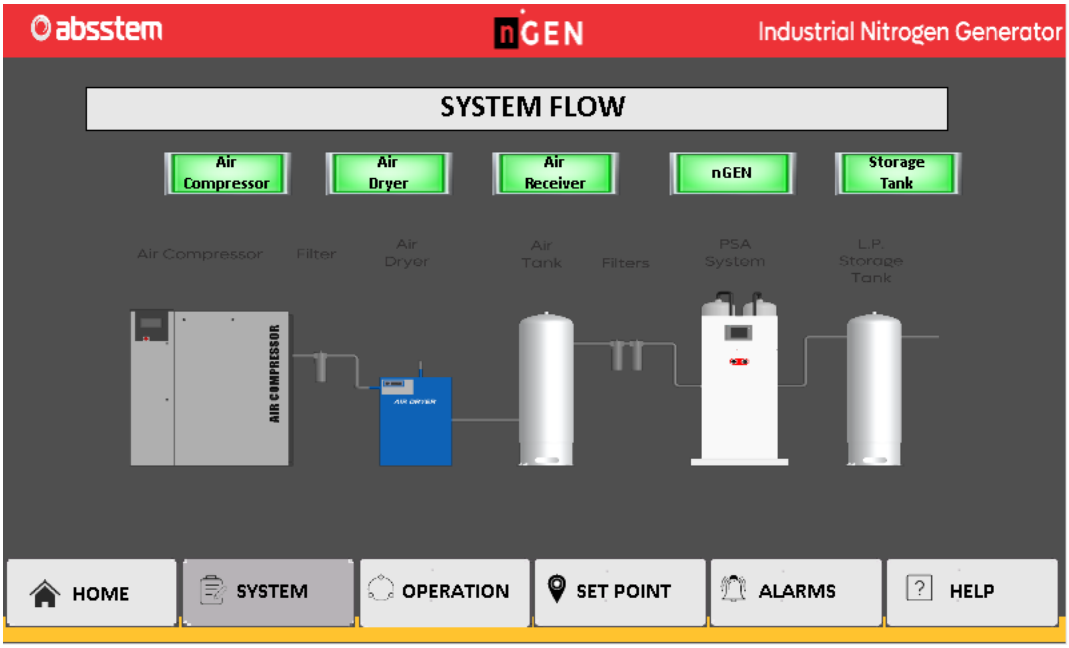
- Low inlet Air pressure
- Low storage pressure
- Purity Low alarm
- Dew point alarm
- Other customizable alarms

Operation Screen:



- **Live valve operation status:** View real-time on/off status of individual valves.
- **Detailed information:** Access valve operation history, set points, and diagnostic data.

System Screen:



- **Visual diagram:** Displays interconnected series of plants for comprehensive system overview.
- **Communication:** Utilize the spare RS485 port for data exchange with other devices or systems.

Additional Features:

- User-friendly interface with customizable layouts and preferences.
- Password protection for secure access control.
- Data logging for historical analysis and trend tracking.
- Remote access capabilities (optional).
- DCS Compatible
- Scada Compatible
- Remote Data Access
- Online Monitoring

Benefits:

- Improved operator Free Operation.
- Enhanced system monitoring and control.
- Reduced downtime and maintenance costs.
- Increased safety and process stability.
- Best Specific Energy as per market.

General features of 'Eye touch' screen

Display section	Display device	TFT color LCD
	Screen size	7" wide screen
	Resolution	WVGA: 800 × 480 dots
	Display size	152.40 (6.00) (W) × 91.44 (3.60) (H) mm (inch)
	Backlight life	Approx. 50000 h (operating ambient temperature: 25 °C, display intensity: 50%
Battery	Life	GT11-50BAT lithium battery
Built-in interface	RS-232	1 channel, transmission speed: 115200, 57600, 38400, 19200, 9600, 4800 bps Connector shape: D-sub 9-pin (male)
	RS-422/485	1 channel, transmission speed: 115200, 57600, 38400, 19200, 9600, 4800 bps Connector shape: D-sub 9-pin (female) Terminating resistor: 330 Ω, 100 Ω, OPEN (Selectable by the terminating resistor setting switch.)
	Ethernet	1 channel Data transfer method: 100BASE-TX, 10BASE-T Connector shape: RJ45 (modular jack) AUTO MDI/MDI-X
	USB (Host)	USB1.1 (Full-Speed 12 Mbps), Connector shape: USB Mini-B
	SD card	1 channel, SDHC compliant (maximum 32 GB)
External dimensions		189 (7.44) (W) × 142 (5.59) (H) × 48 (1.89) (D) mm (inch)
Weight (excluding a fitting)		0.7 (1.5) kg (lb)

Note: This document provides a general overview. Specific features and functionalities may vary depending on your plant configuration and specific model of HMI.

I hope this information is helpful.

Operating atmosphere *6 No greasy fumes, corrosive gas, flammable gas, excessive conductive dust, and direct sunlight (as well as at storage), Operating altitude *3 2000 m or less Grounding with a ground resistance of 100 Ω or less by using a ground cable that has a cross-sectional area of 2 mm2 or more. If impossible, connect the ground cable to the control panel. Protective structure-Front: IP67F *6*8 In control panel: IP2X.